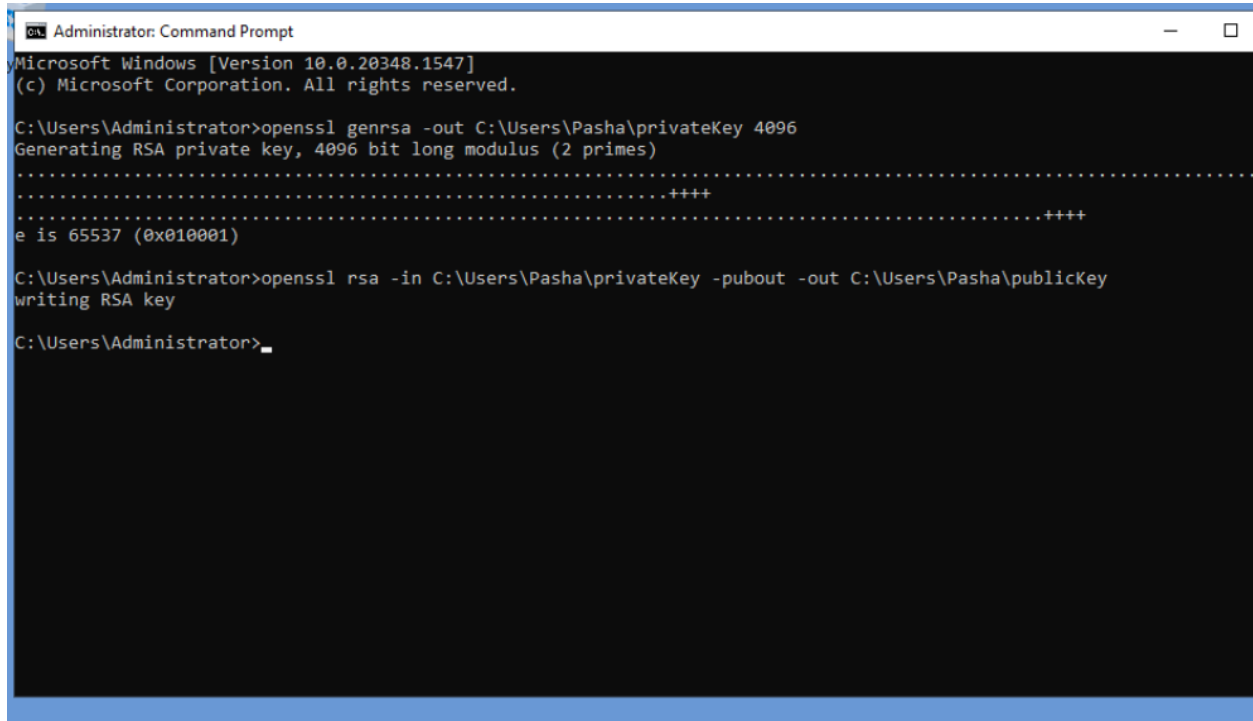


Lab 1.1

Step 1. Create Pasha's RSA key pairs on ZYWIN01



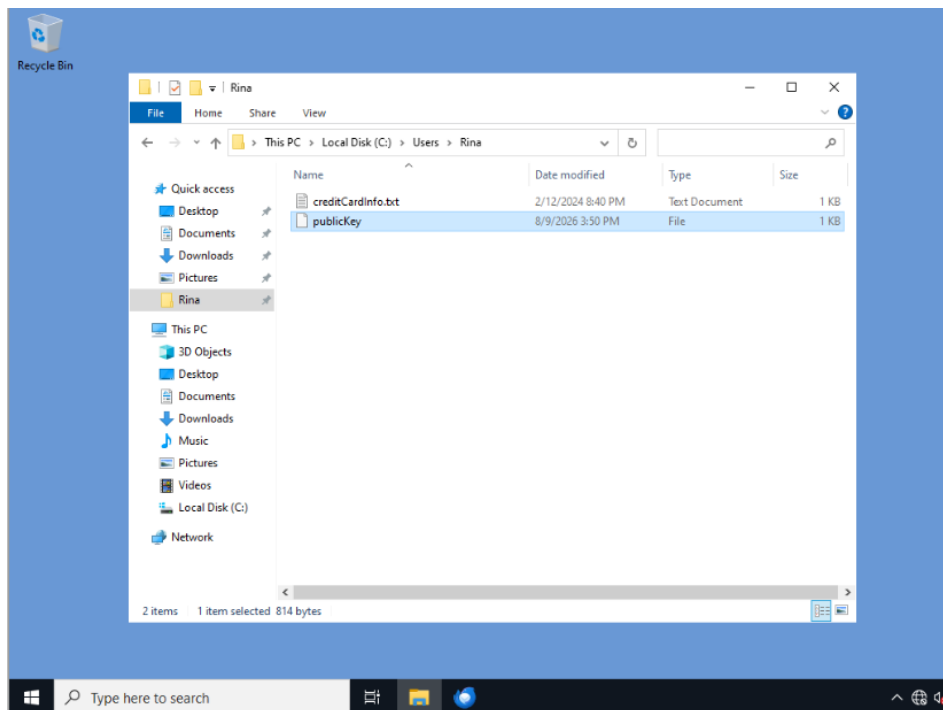
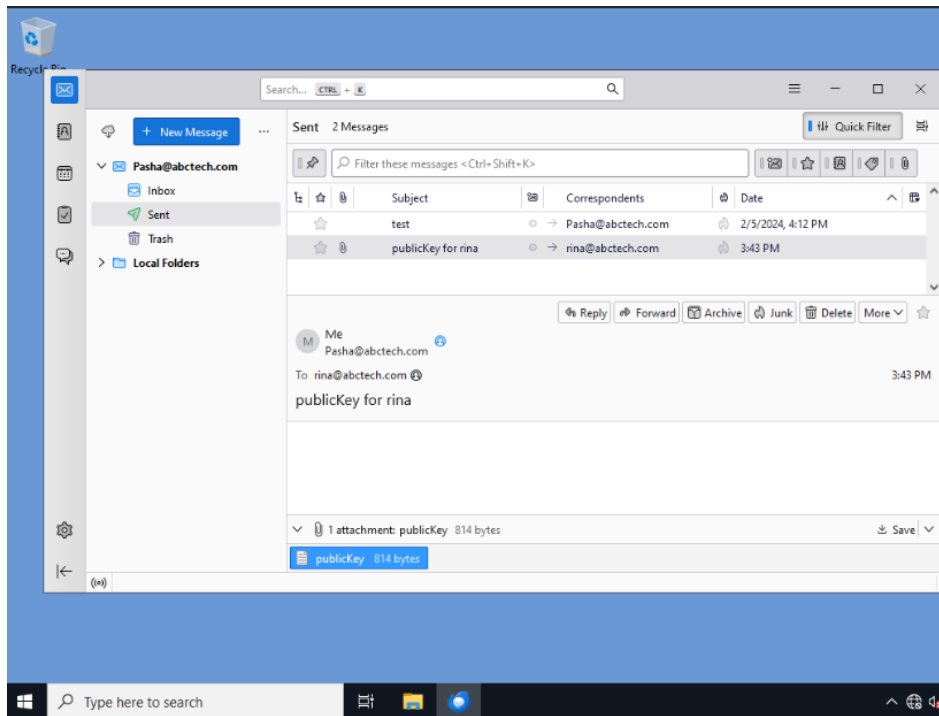
```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.20348.1547]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>openssl genrsa -out C:\Users\Pasha\privateKey 4096
Generating RSA private key, 4096 bit long modulus (2 primes)
.....+++++
.....+++++
e is 65537 (0x010001)

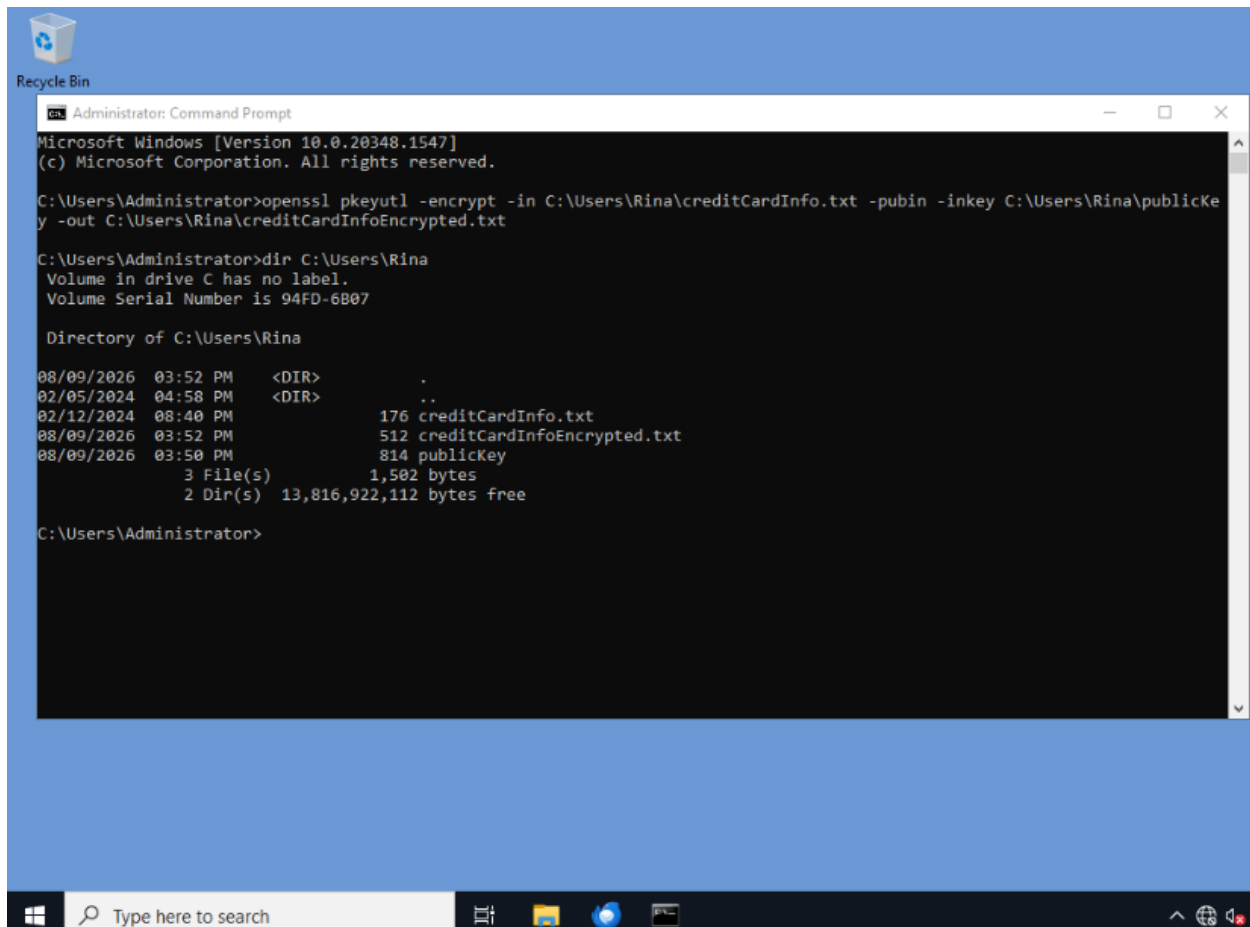
C:\Users\Administrator>openssl rsa -in C:\Users\Pasha\privateKey -pubout -out C:\Users\Pasha\publicKey
writing RSA key

C:\Users\Administrator>
```

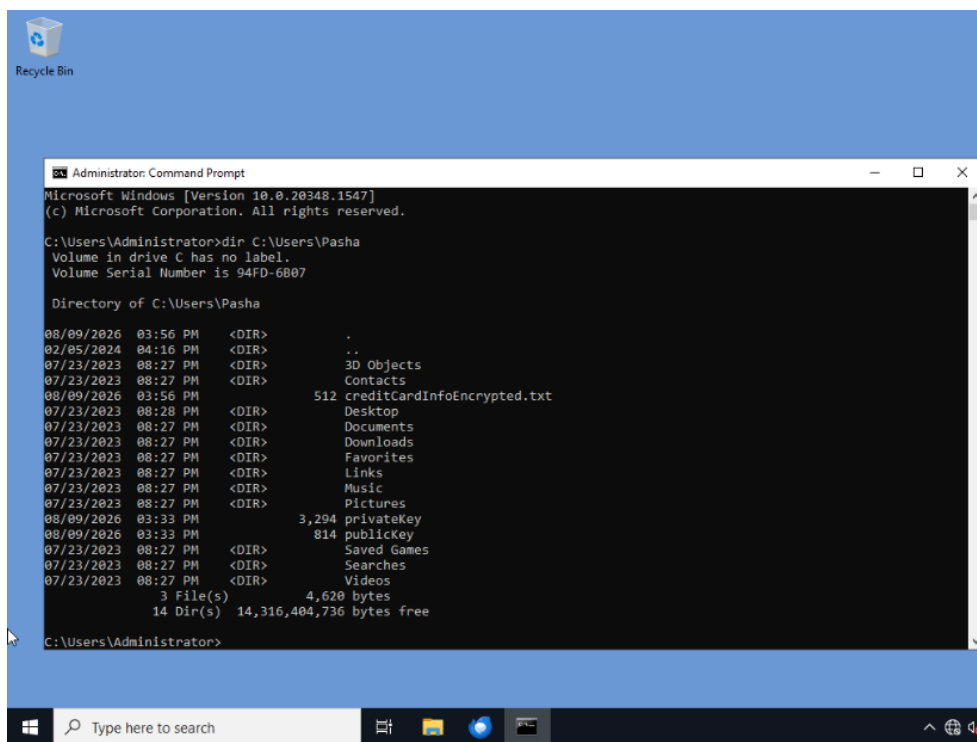
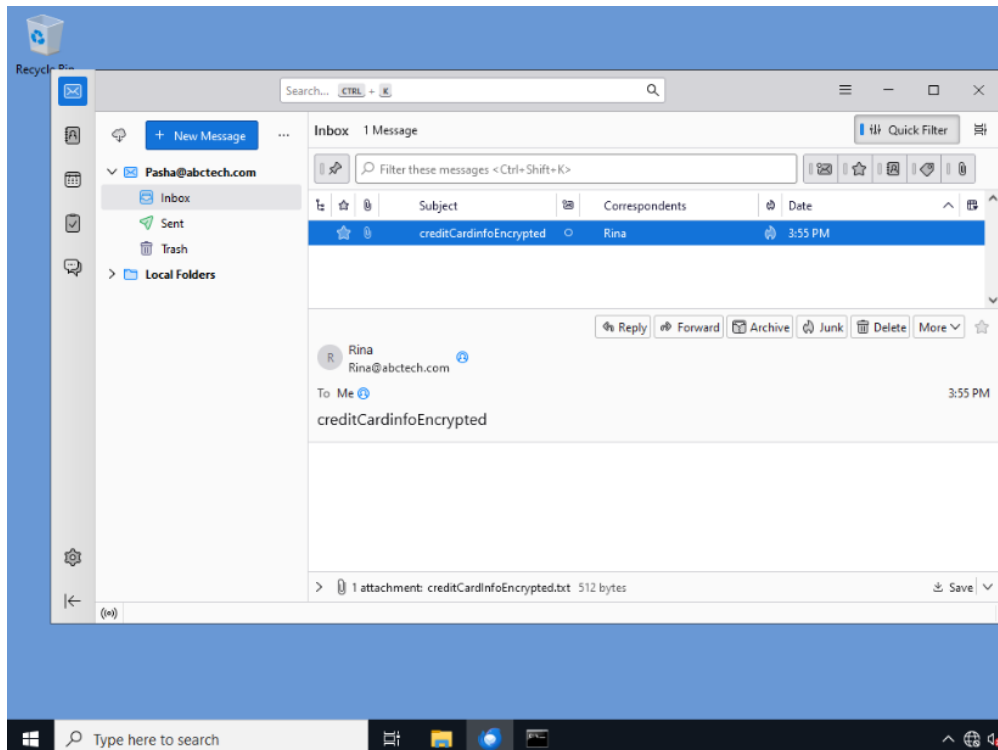
Step 2. Share Pasha's public key with Rina via email, saving the key in C:\Users\Rina on ZYWIN02



Step 3. Encrypt C:\Users\Rina\creditCardInfo.txt with Pasha's public key, saving the encrypted version as creditCardInfoEncrypted.txt



Step 4. Email creditCardInfoEncrypted.txt to Pasha, saving the file in C:\Users\Pasha



Step 5. Decrypt creditCardInfoEncrypted.txt, saving the decrypted version as creditCardInfo.txt

```
Administrator: Command Prompt
07/23/2023 08:27 PM <DIR> Searches
07/23/2023 08:27 PM <DIR> Videos
3 File(s) 4,620 bytes
14 Dir(s) 14,316,404,736 bytes free

C:\Users\Administrator>openssl pkeyutl -decrypt -in C:\Users\Pasha\creditCardInfoEncrypted.txt -inkey C:\Users\Pasha\privateKey -out C:\Users\Pasha\creditCardInfo.txt

C:\Users\Administrator>type C:\Users\Pasha\creditCardInfo.txt
Name SSN Credit Card Number
Robert Aragon 489-36-8350 4929-3813-3266-4295
Ashley Borden 514-14-8905 5370-4638-8881-3020
Thomas Conley 690-05-5315 4916-4811-5814-8111

C:\Users\Administrator>dir C:\Users\Pasha
Volume in drive C has no label.
Volume Serial Number is 94FD-6807

Directory of C:\Users\Pasha

08/09/2026 04:00 PM <DIR> .
02/05/2024 04:16 PM <DIR> ..
07/23/2023 08:27 PM <DIR> 3D Objects
07/23/2023 08:27 PM <DIR> Contacts
08/09/2026 04:00 PM 176 creditCardInfo.txt
08/09/2026 03:56 PM 512 creditCardInfoEncrypted.txt
07/23/2023 08:28 PM <DIR> Desktop
07/23/2023 08:27 PM <DIR> Documents
07/23/2023 08:27 PM <DIR> Downloads
07/23/2023 08:27 PM <DIR> Favorites
07/23/2023 08:27 PM <DIR> Links
07/23/2023 08:27 PM <DIR> Music
07/23/2023 08:27 PM <DIR> Pictures
08/09/2026 03:33 PM 3,294 privateKey
08/09/2026 03:33 PM 814 publicKey
07/23/2023 08:27 PM <DIR> Saved Games
07/23/2023 08:27 PM <DIR> Searches
07/23/2023 08:27 PM <DIR> Videos
4 File(s) 4,796 bytes
14 Dir(s) 14,314,881,024 bytes free

C:\Users\Administrator>
```

Step 6. Once the lab is completed, click Delete VMs in the right pane.

The image shows a lab completion interface on the left and a virtual machine console on the right.

Lab Completion Interface (Left):

- 1. Create Pasha's RSA key pairs on ZYWIN01.
 - 1.1 RSA private key with 4096-bits created at **C:\Users\Pasha\privateKey**. Current VMs: 2/2, Done. ☒ [Feedback?](#)
 - 1.2 RSA public key created at **C:\Users\Pasha\publicKey**. Current VMs: 2/2, Done. ☒ [Feedback?](#)
- 2. Share Pasha's public key with Rina via email, saving the key in **C:\Users\Rina** on ZYWIN02.

Pasha's public key saved in **C:\Users\Rina**. Current VMs: 4/4, Done. ☒ [Feedback?](#)
- 3. Encrypt **C:\Users\Rina\creditCardInfo.txt** with Pasha's public key, saving the encrypted version as **creditCardInfoEncrypted.txt**.

creditCardInfoEncrypted.txt saved in **C:\Users\Rina**. Current VMs: 4/4, Done. ☒ [Feedback?](#)
- 4. Email **creditCardInfoEncrypted.txt** to Pasha, saving the file in **C:\Users\Pasha**.

creditCardInfoEncrypted.txt saved in **C:\Users\Pasha**. Current VMs: 4/4, Done. ☒ [Feedback?](#)
- 5. Decrypt **creditCardInfoEncrypted.txt**, saving the decrypted version as **creditCardInfo.txt**.

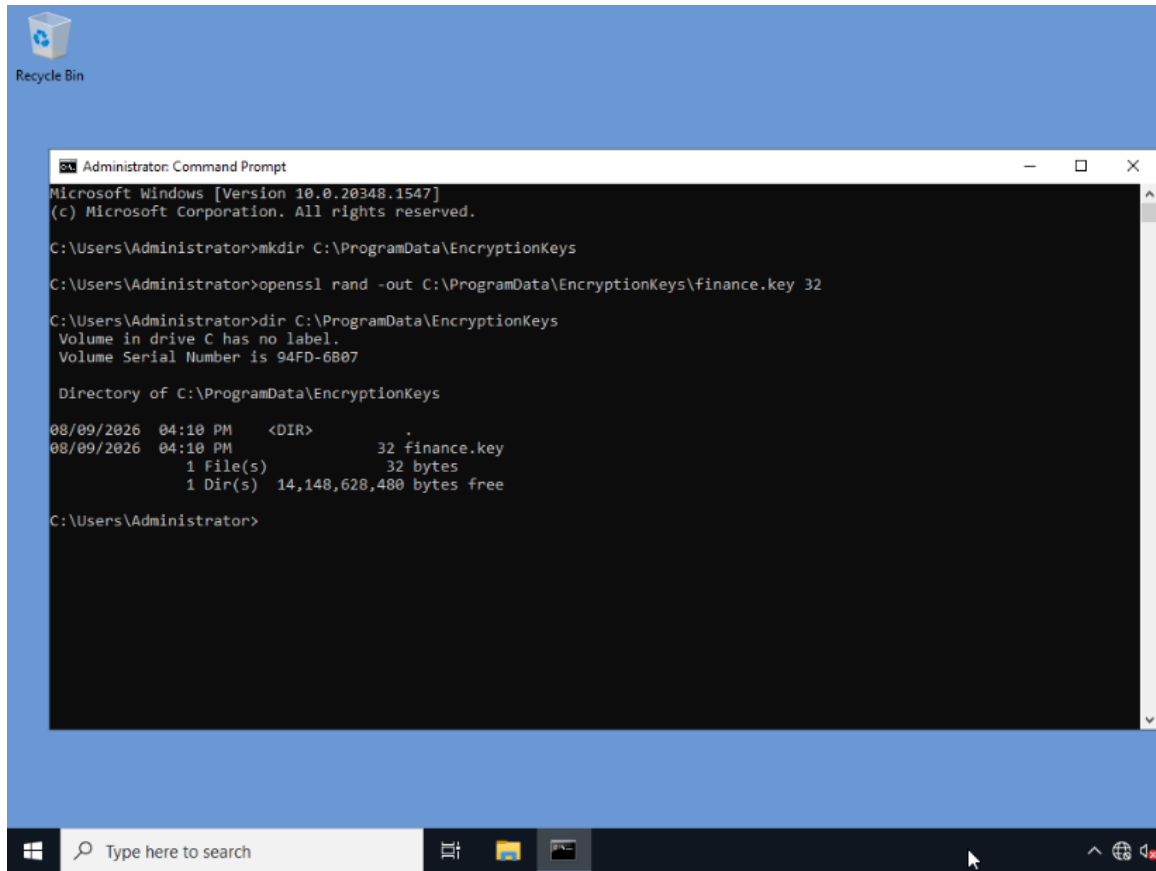
creditCardInfo.txt saved at **C:\Users\Pasha**. Current VMs: 4/4, Done. ☒ [Feedback?](#)
- 6. Once the lab is completed, click **Delete VMs** in the right pane.

☒ You have completed all the goals for this lab.

Delete VMs?
Are you sure you want to delete the VMs?

Lab 1.2

Step 1. Generate a random key for use with AES-256-CBC encryption on ZYWIN01



Step 2. Securely transfer finance.key using SCP to ZYWIN02



Recycle Bin

```
Administrator: Command Prompt
\nfinance.key
The authenticity of host 'zywin02 (fe80::49c:3a1d:731:eca1%6)' can't be established.
ECDSA key fingerprint is SHA256:6TIRATU1Owqa/mowlyH4OaYMBH23R4Nv8TEgM0Ep4AY.
Are you sure you want to continue connecting (yes/no/[fingerprint])?
Warning: Permanently added 'zywin02,fe80::49c:3a1d:731:eca1%6' (ECDSA) to the list of known hosts.
Administrator@zywin02's password:
C:/ProgramData/EncryptionKeys/finance.key: No such file or directory

C:\Users\Administrator>dir C:\ProgramData\EncryptionKeys
Volume in drive C has no label.
Volume Serial Number is 94FD-6807

Directory of C:\ProgramData

File Not Found

C:\Users\Administrator>mkdir C:\ProgramData\EncryptionKeys

C:\Users\Administrator>dir C:\ProgramData\EncryptionKeys
Volume in drive C has no label.
Volume Serial Number is 94FD-6807

Directory of C:\ProgramData\EncryptionKeys

08/09/2026  04:15 PM    <DIR>          .
08/09/2026  04:15 PM                32 finance.key
               1 File(s)                32 bytes
               1 Dir(s) 13,700,247,552 bytes free

C:\Users\Administrator>
```



Recycle Bin

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.20348.1547]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>mkdir C:\ProgramData\EncryptionKeys

C:\Users\Administrator>openssl rand -out C:\ProgramData\EncryptionKeys\finance.key 32

C:\Users\Administrator>dir C:\ProgramData\EncryptionKeys
Volume in drive C has no label.
Volume Serial Number is 94FD-6807

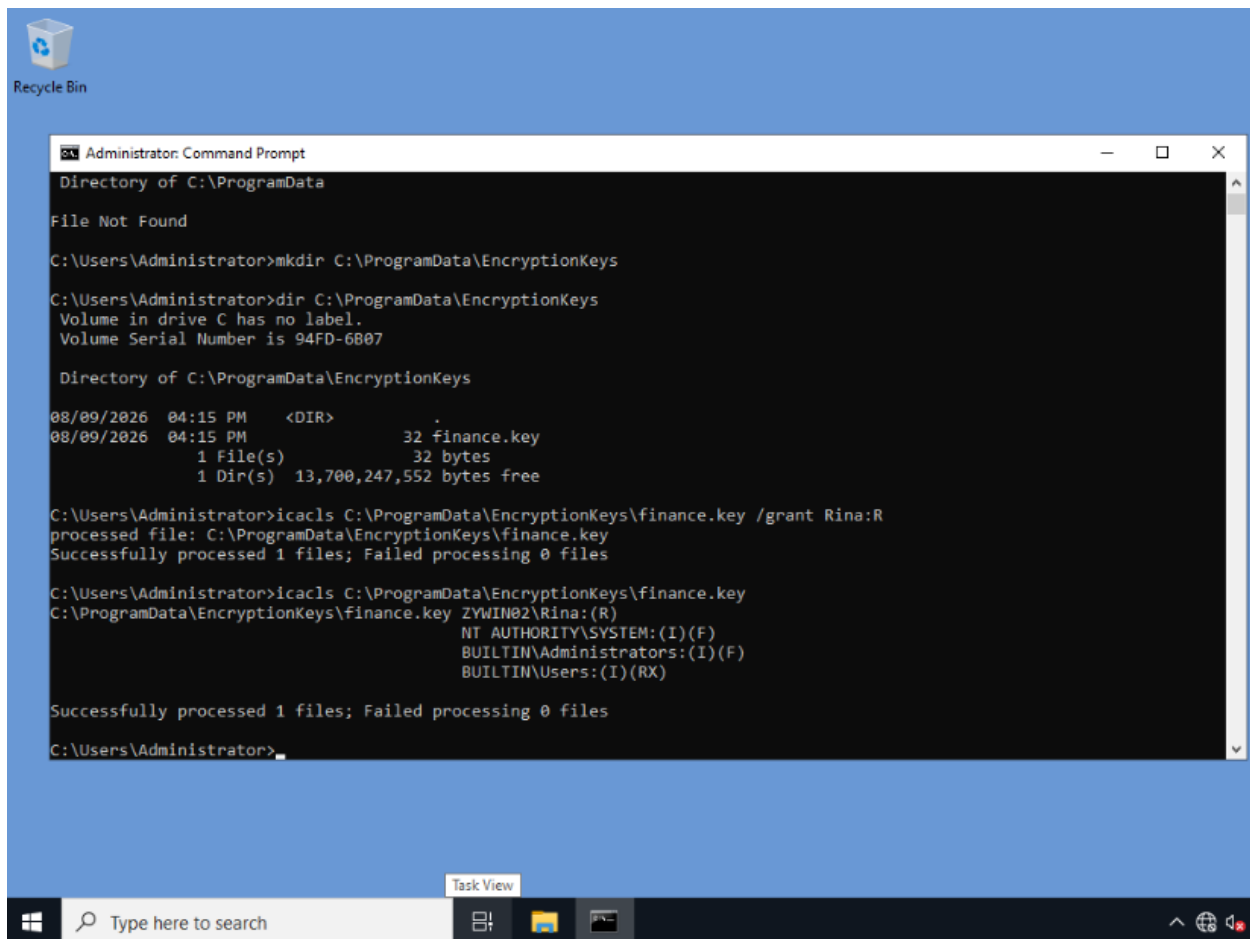
Directory of C:\ProgramData\EncryptionKeys

08/09/2026  04:10 PM    <DIR>          .
08/09/2026  04:10 PM                32 finance.key
               1 File(s)                32 bytes
               1 Dir(s) 14,148,628,480 bytes free

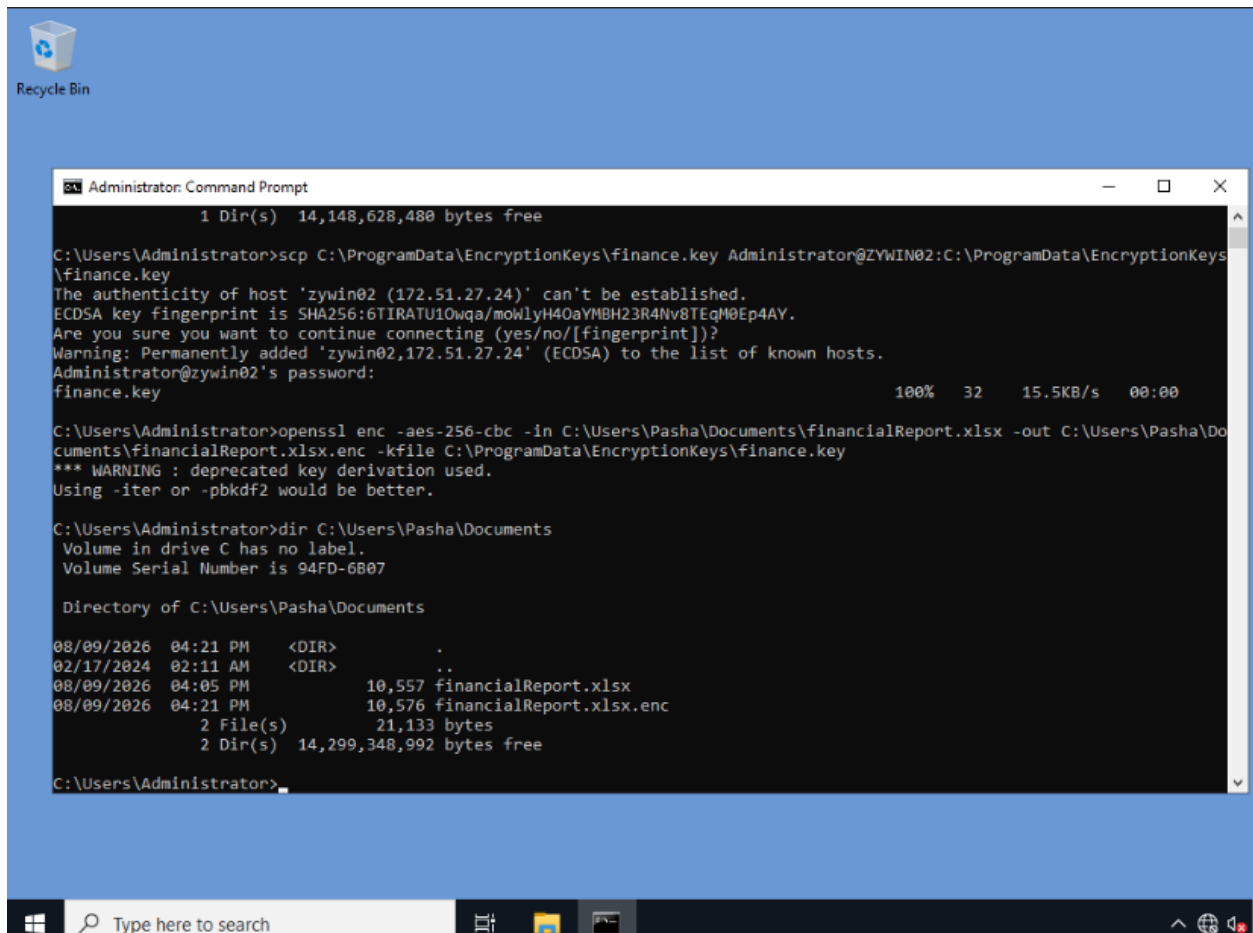
C:\Users\Administrator>scp C:\ProgramData\EncryptionKeys\finance.key Administrator@ZYWIN02:C:\ProgramData\EncryptionKeys
\nfinance.key
The authenticity of host 'zywin02 (172.51.27.24)' can't be established.
ECDSA key fingerprint is SHA256:6TIRATU1Owqa/mowlyH4OaYMBH23R4Nv8TEgM0Ep4AY.
Are you sure you want to continue connecting (yes/no/[fingerprint])?
Warning: Permanently added 'zywin02,172.51.27.24' (ECDSA) to the list of known hosts.
Administrator@zywin02's password:
finance.key                                     100% 32  15.5KB/s  00:00

C:\Users\Administrator>
```

Step 3. Grant read-only permission to Rina for the finance.key on ZYWIN02



Step 4. Encrypt C:\Users\Pasha\Documents\financialReport.xlsx using finance.key, saving the encrypted version as C:\Users\Pasha\Documents\financialReport.xlsx.enc on ZYWIN01



```
Administrator: Command Prompt

1 Dir(s) 14,148,628,480 bytes free

C:\Users\Administrator>scp C:\ProgramData\EncryptionKeys\finance.key Administrator@ZYWIN02:C:\ProgramData\EncryptionKeys\finance.key
The authenticity of host 'zywin02 (172.51.27.24)' can't be established.
ECDSA key fingerprint is SHA256:6TIRATU1Owqa/mowlyH4OaYMBH23R4Nv8TEqM0Ep4AY.
Are you sure you want to continue connecting (yes/no/[fingerprint])?
Warning: Permanently added 'zywin02,172.51.27.24' (ECDSA) to the list of known hosts.
Administrator@zywin02's password:
finance.key
100% 32 15.5KB/s 00:00

C:\Users\Administrator>openssl enc -aes-256-cbc -in C:\Users\Pasha\Documents\financialReport.xlsx -out C:\Users\Pasha\Documents\financialReport.xlsx.enc -kfile C:\ProgramData\EncryptionKeys\finance.key
*** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.

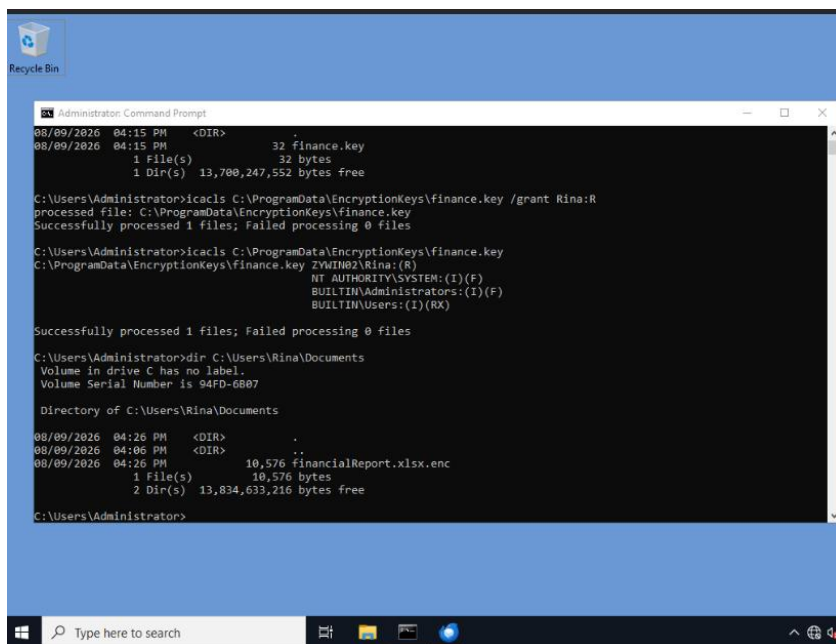
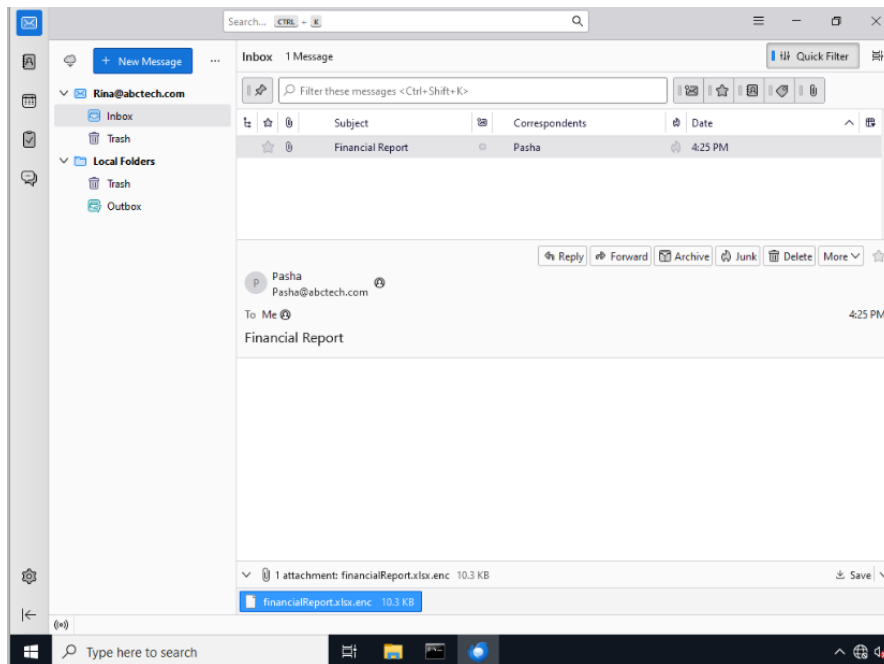
C:\Users\Administrator>dir C:\Users\Pasha\Documents
Volume in drive C has no label.
Volume Serial Number is 94FD-6B07

Directory of C:\Users\Pasha\Documents

08/09/2026 04:21 PM <DIR> .
02/17/2024 02:11 AM <DIR> ..
08/09/2026 04:05 PM 10,557 financialReport.xlsx
08/09/2026 04:21 PM 10,576 financialReport.xlsx.enc
2 File(s) 21,133 bytes
2 Dir(s) 14,299,348,992 bytes free

C:\Users\Administrator>
```

Step 5. Email financialReport.xlsx.enc to Rina, saving the file in C:\Users\Rina\Documents on ZYWIN02



Step 6. Decrypt financialReport.xlsx.enc using finance.key, saving the decrypted file as financialReport.xlsx on ZYWIN02

```
Administrator: Command Prompt
processed file: C:\ProgramData\EncryptionKeys\finance.key
Successfully processed 1 files; Failed processing 0 files

C:\Users\Administrator>icacls C:\ProgramData\EncryptionKeys\finance.key
C:\ProgramData\EncryptionKeys\finance.key ZYWIN02\Rina:(R)
NT AUTHORITY\SYSTEM:(I)(F)
BUILTIN\Administrators:(I)(F)
BUILTIN\Users:(I)(RX)

Successfully processed 1 files; Failed processing 0 files

C:\Users\Administrator>dir C:\Users\Rina\Documents
Volume in drive C has no label.
Volume Serial Number is 94FD-6B07

Directory of C:\Users\Rina\Documents

08/09/2026  04:26 PM    <DIR>          .
08/09/2026  04:06 PM    <DIR>          ..
08/09/2026  04:26 PM                10,576 financialReport.xlsx.enc
               1 File(s)              10,576 bytes
               2 Dir(s)  13,834,633,216 bytes free

C:\Users\Administrator>openssl enc -d -aes-256-cbc -in C:\Users\Rina\Documents\financialReport.xlsx.enc -out C:\Users\Rina\Documents\financialReport.xlsx -kfile C:\ProgramData\EncryptionKeys\finance.key
*** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.

C:\Users\Administrator>dir C:\Users\Rina\Documents
Volume in drive C has no label.
Volume Serial Number is 94FD-6B07

Directory of C:\Users\Rina\Documents

08/09/2026  04:29 PM    <DIR>          .
08/09/2026  04:06 PM    <DIR>          ..
08/09/2026  04:29 PM                10,557 financialReport.xlsx
08/09/2026  04:26 PM                10,576 financialReport.xlsx.enc
               2 File(s)              21,133 bytes
               2 Dir(s)  13,816,913,920 bytes free

C:\Users\Administrator>_
```

Step 7. Once the lab is completed, click Delete VMs in the right pane.

The screenshot displays a virtual machine lab interface. On the left, a list of tasks is shown, all marked as completed with green checkmarks and 'Done' status. The tasks include transferring a finance key, granting permissions, encrypting a file, emailing it, and decrypting it. A green checkmark at the bottom indicates that all goals for the lab have been completed.

On the right, the 'Delete VMs' button is highlighted. A dialog box titled 'Delete VMs?' is open, asking 'Are you sure you want to delete the VMs?'. The dialog has two buttons: 'No, keep the VMs' and 'Delete VMs'.

The background shows a terminal window with the following commands and output:

```
Administrator Command Prompt
C:\ProgramData\EncryptionKeys>finance.key
Successfully processed 1 files; failed processing 0 files

C:\Users\Administrator>icacls C:\ProgramData\EncryptionKeys\finance.key
C:\ProgramData\EncryptionKeys\finance.key ZYWIN02\Rina:(O)
NT AUTHORITY\SYSTEM:(I)(F)
BUILTIN\Administrators:(I)(F)
BUILTIN\Users:(I)(RX)

Successfully processed 1 files; failed processing 0 files

C:\Users\Administrator>dir C:\Users\Rina\Documents
Volume in drive C: has no label.
Volume Serial Number is 94FD-6867

Directory of C:\Users\Rina\Documents

08/09/2025 04:29 PM <DIR> .
08/09/2025 04:29 PM <DIR> ..
08/09/2025 04:29 PM 18,557 financialReport.xlsx
08/09/2025 04:29 PM 18,576 financialReport.xlsx.enc
2 File(s) 21,133 bytes
2 Dir(s) 13,816,913,920 bytes free

C:\Users\Administrator>
```